

Question:
Find the number of ways to place a digit in each cell of a 2x3 grid so that the sum of the two numbers formed by reading left to right is \$999\$, and the sum of the three numbers formed by reading top to bottom is \$99\$. The grid below is an example of such an arrangement because \$8+991=999\$ and \$9+9+81=99\$.

$$\begin{array}{|c|c|c|} \hline 0 & 0 & 8 \\ \hline 9 & 9 & 1 \\ \hline \end{array}$$

Reasoning + Answer:

<think>

1 . We need to find the number of ways to place digits in a 2 x 3 grid such that the sum of the two numbers formed by reading left to right is 999 , and the sum of the three numbers formed by reading top to bottom is 99 .

2 . Let's denote the grid as :

$$\begin{array}{|c|c|c|} \hline a & b & c \\ \hline d & e & f \\ \hline \end{array}$$

3 . According to the problem , we have the following conditions :

- The sum of the two numbers formed by reading left to right is 999 .
- The sum of the three numbers formed by reading top to bottom is 99 .

4 . Let's break this down :

- For the sum of the numbers formed by reading left to right :

$$10b + c + 10a + b = 11a + 11b + c = 999$$

- For the sum of the numbers formed by reading top to bottom :

$$\begin{aligned} a + d + f &= 9 \\ b + e + f &= 9 \\ c + d &= 9 \end{aligned}$$

5 . Let's use the condition $(a + b + c = 99)$ from the left-to-right sum .

6 . Also , we have :

$$\begin{aligned} a + c + d &= 9 \quad \text{(top-to-bottom left column sum)} \\ b + c + f &= 9 \quad \text{(top-to-bottom middle column sum)} \\ d + f &= 9 \quad \text{(top-to-bottom right column sum)} \end{aligned}$$

7 . We now have two separate sums we can solve :

$$\begin{aligned} a + c + d &= 9 \\ c + d + f &= 9 \end{aligned}$$

8 . Substituting $(c + d)$ from the left-to-right sum :

$$c + d = 99 - (a + b)$$

9 . Thus , substituting back into the sums :

- Summing the first two :

$$a + (99 - (a + b)) + (99 - (a + b)) = 99$$

10 . Solve :

- Check conditions :
- Check $(a + c + d)$:

$$a + (99 - (a + b)) + (99 - (a + b)) = 99$$

- Checking middle constraints :

- $((a + b) \text{ to })$

- Final set up known digits :

2 . The solution specifics under diaphanous checks constraint specific product utations exhaustives no worse reason the final # valid

</think>

<answer> 3 </answer>